

Technical Assessment

Mini Kanban Board — Full-Stack Engineering Challenge

Role: **Full-Stack Engineer** Timeframe: **4 Days**

Tech Stack

Frontend

Next.js, React (TypeScript preferred), Tailwind CSS

Backend

NestJS (preferred) or Node.js / Express.js with TypeScript

Database

PostgreSQL with Prisma (preferred)

DevOps

Docker (preferable)

Project Overview

Build a functional **Mini Kanban Board** application where users can create boards, organize workflow columns, and manage tasks. You are expected to design your own **database schema** and **system architecture** to handle collaboration, access permissions, and drag-and-drop task reordering.

Core Requirements

1. Authentication & Collaboration

- User registration and login with **token-based authentication**.
- **Board Sharing**: Boards must have an owner and allow sharing access with other registered users.
- **Access Control**: Enforce authorization rules so users can only view or mutate boards, columns, and tasks they have explicit access to. Prevent unauthorized cross-board access.

2. Workflow Management & Task Movement

- Full management of **Boards, Columns, and Tasks**.
- **Task Movement API**: Implement an endpoint to handle moving tasks:
 - Reordering tasks within the same column.
 - Moving a task across different columns to a specific position index.
- **Order Consistency**: Ensure task ordering remains stable, accurate, and conflict-free when tasks are rearranged.

3. Frontend

- Interactive board view supporting **drag-and-drop** task movement.

Submission & Deliverables

- **Single Repository**: Submit a single GitHub repository containing both frontend and backend directories.
- **Setup Instructions**: Include a `README.md` with step-by-step local setup instructions and sample environment variables.
- **Docker (Preferable)**: A `docker-compose.yml` to spin up the database and services locally with minimal setup.
- **Deployment (Optional)**: If possible, provide a link to a live deployed version of the application.